

Developing a **React Redux** Application Using create-react-app

This article explains the development of a React application, using Redux for state management, interfacing with the Express framework for data and using create-react-app as the frontend build pipeline.



React is a JavaScript library for building user interfaces, and can be used to build single page applications. It is focused on creating the V of MVC, i.e., the View layer. Using a concept called Virtual DOM, components can be rendered with a good performance. There is large amount of state that needs to be managed by single page applications. The state includes server responses, cached and locally created data, and various UI states. Redux is a popular framework for managing the state. The data displayed by the client side UI can be provided from a back-end server. Express is a commonly used Web framework based on Node.js. This article explains the development of a React application using Redux for state management and interfacing with the Express framework for data, along with using the create-react-app for setting up the development environment. The create-react-app seamlessly handles the front-end build pipeline, enabling us to focus on writing the application logic.

To get an understanding of this article, readers should know the basics of React, Redux, Express.js, and Node.js.

Contacts application

The concepts of developing an application using React, Redux and Express will be illustrated through a Contacts application. We will keep the application simple, as the main objective is to illustrate the various concepts in an easy to

understand manner. Using the application, we will be able to view the Contacts list, add a new contact and delete a contact. We will also use redux-thunk, a middleware, to perform the action asynchronously.

Setting up the development environment

We will develop the application using Windows as the development machine.

1. Install Node.js from <https://nodejs.org>. Download the 32-bit or 64-bit binary depending upon the architecture of the system.
2. Install create-react-app, using which we will start building the Contacts application.
3. Generate the basic code template, which we will modify to add the features required for the Contacts application.

```
npm install -g create-react-app
```

```
create-react-app contacts
```

This will create a directory called *contacts* in the current folder. Inside that folder, we can see an initial project structure generated; the required dependencies are automatically installed.